

Predicting Financial Inclusion in Africa

Using machine learning to predict bank-account ownership

Kenya · Rwanda · Tanzania · Uganda

AGENDA

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2. A DATA OVERVIEW
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BACKGROUND

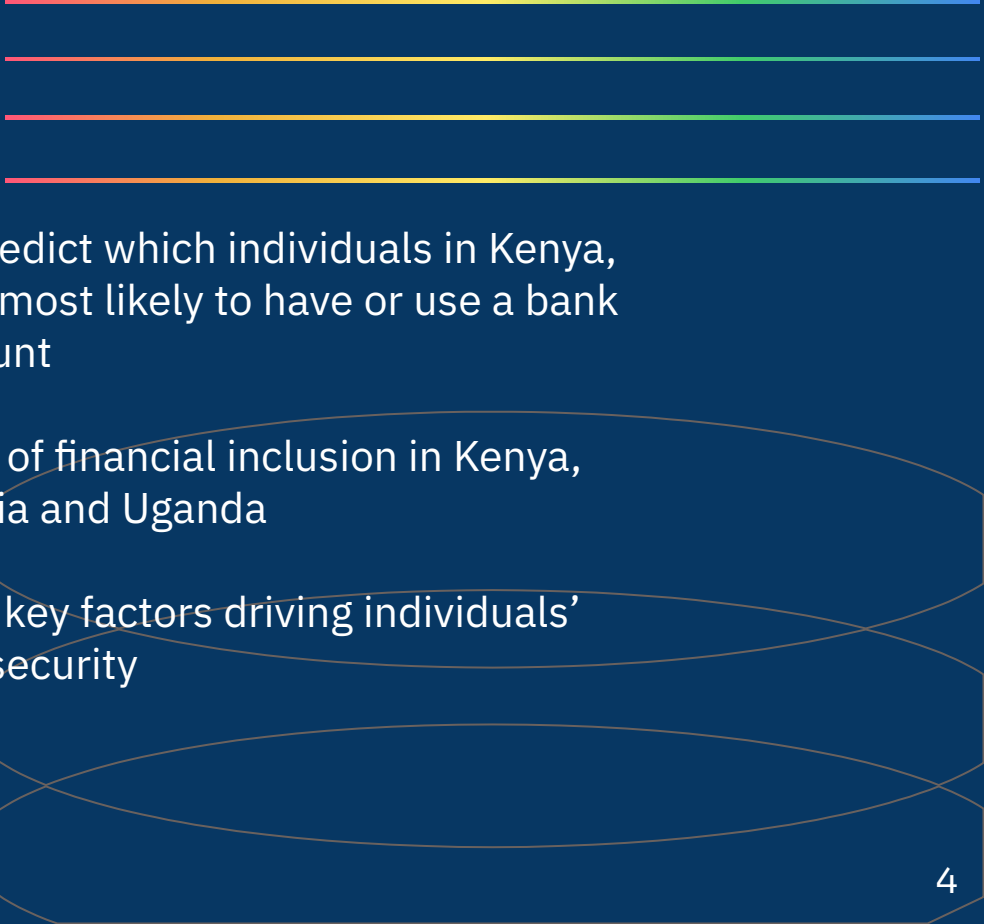
Financial inclusion is a main obstacle to economic and human development in Africa

Financial inclusion could be defined as saving, ability to make payments, helping businesses to build up credit-worthiness which improves access to loans, insurance, and other related services

Bank account access is regarded as a main indicator of Financial Inclusion

Access to bank accounts is an essential contributor to long-term economic growth

OUR TASK



1. Use machine learning models to predict which individuals in Kenya, Rwanda, Tanzania and Uganda are most likely to have or use a bank account
2. Provide an indication of the state of financial inclusion in Kenya, Rwanda, Tanzania and Uganda
3. Provide insights into some of the key factors driving individuals' financial security

A DATA OVERVIEW



23,524

respondents

11

model features

3,312 with
an account

20,212
without an
account

Geography

- Country
- Survey year
- Urban or rural

Household and demographics

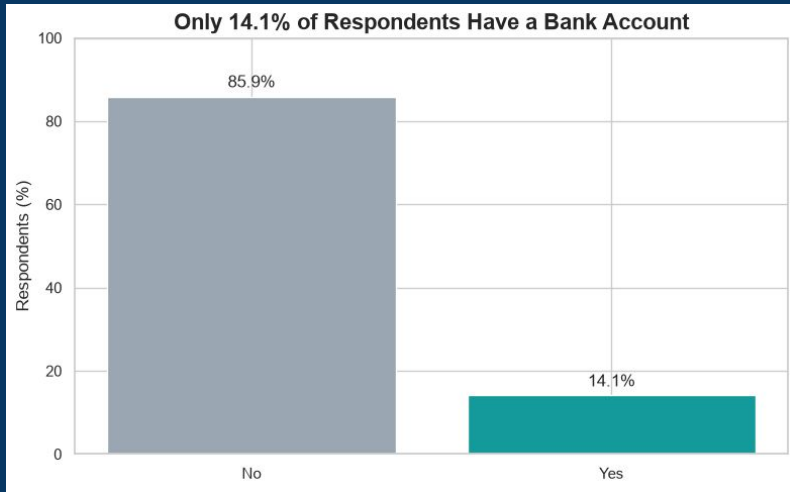
- Age
- Gender
- Household size
- Relationship to head
- Marital status

Access and opportunity

- Mobile-phone access
- Education level
- Employment type

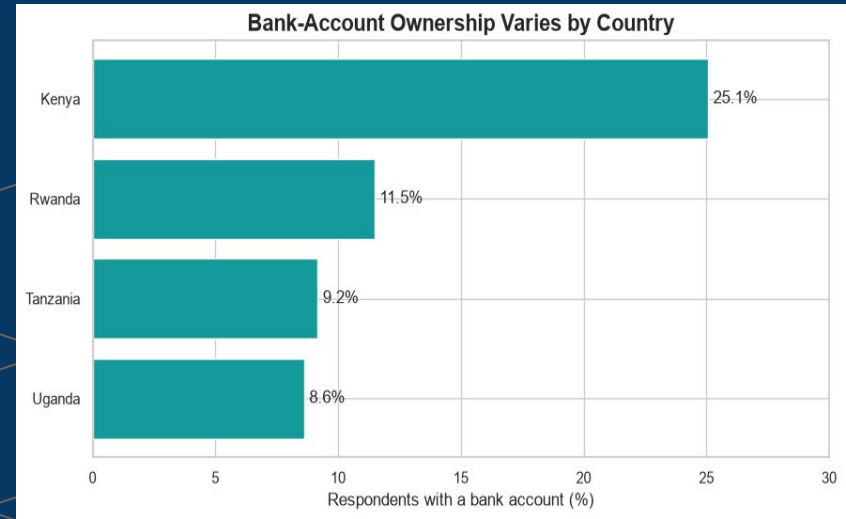
note: bank_account and uniqueid are not included here.

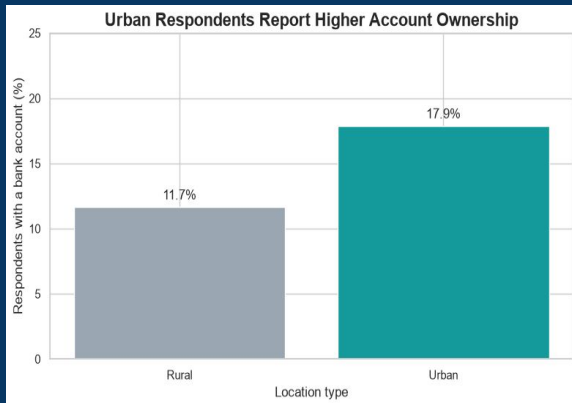
uniqueid is only an identifier and not information that the model can learn from.
bank_account is our target. It is separated for our Machine Learning model.



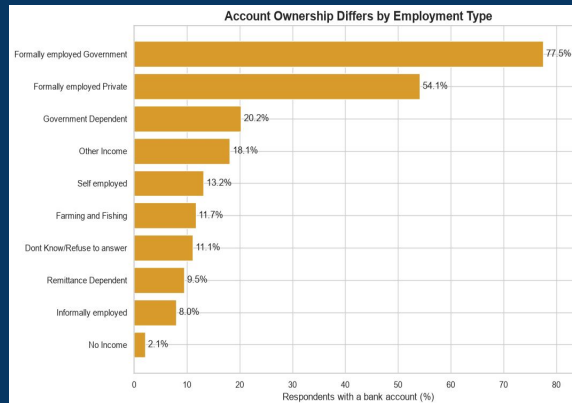
The target variable is strongly imbalanced. Approximately 86% of respondents do not have a bank account, while only 14% do. This means a model could achieve high overall Accuracy while still performing poorly for the smaller group.

Kenya has the highest observed bank-account ownership rate at 25.1%. Rwanda, Tanzania, and Uganda have considerably lower rates. This suggests that financial-inclusion strategies should be adapted to national and local conditions

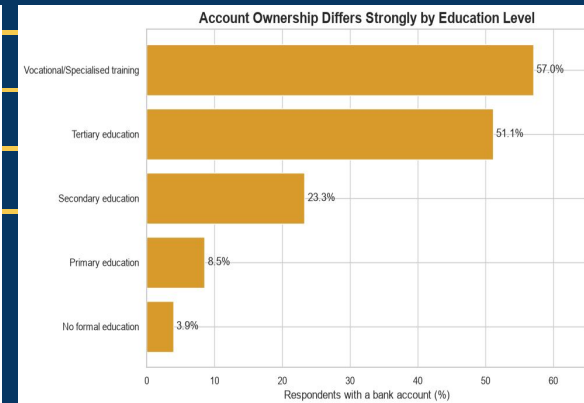




Urban respondents are more likely to have a bank account, suggesting that rural populations may face greater barriers to accessing formal financial services.



Employment type is strongly associated with bank-account ownership. Formally employed respondents are much more likely to have an account, while financially vulnerable and informally employed groups may face greater barriers to formal banking.



Respondents with vocational or tertiary education reported considerably higher account-ownership rates than respondents with primary or no formal education.

HOW WE USED MACHINE LEARNING

We used the data in 3x different Machine Learning models that will each predict whether or not respondents have a bank account

We used the same process with each model to ensure fair testing

The bank_account field was separated from the training data so the model could predict without influence from these responses

After checking each of the models against one another, we chose the best model based on our selected metrics then fine-tuned it

THE MODEL WE CHOSE



Model	MAE ↓	Accuracy ↑	Recall ↑	Precision ↑
Logistic Regression	11.09%	88.91%	36.10%	70.7%
Random Forest	13.69%	86.31%	42.45%	51.7%
XGBoost	10.90%	89.10%	36.10%	72.6%
Tuned XGBoost	10.90%	89.10%	37.16%	71.7%

We selected the model that made the fewest mistakes overall while still producing the most reliable predictions across the full sample.

Another model found more account holders, but it also made substantially more incorrect predictions, so we treated it as a useful comparison rather than the final choice.

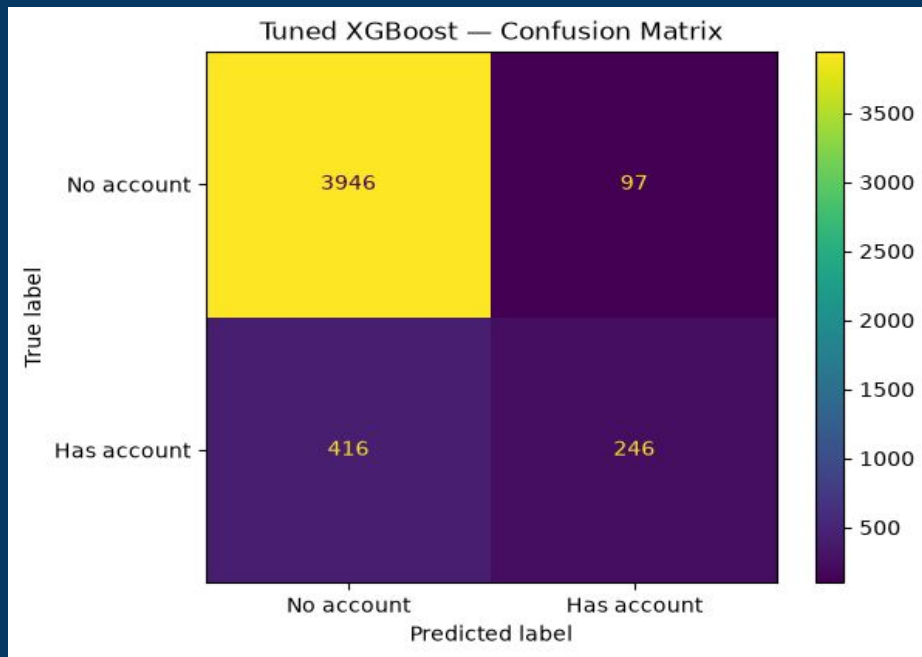
At its current decision threshold, the model is conservative.

When it predicts that someone has a bank account, it is correct about **72% of the time (precision)**.

However, it only identifies about **37% of all actual account holders (recall)***

In other words, it is relatively precise but it misses many true account holders.

OUR FINDINGS

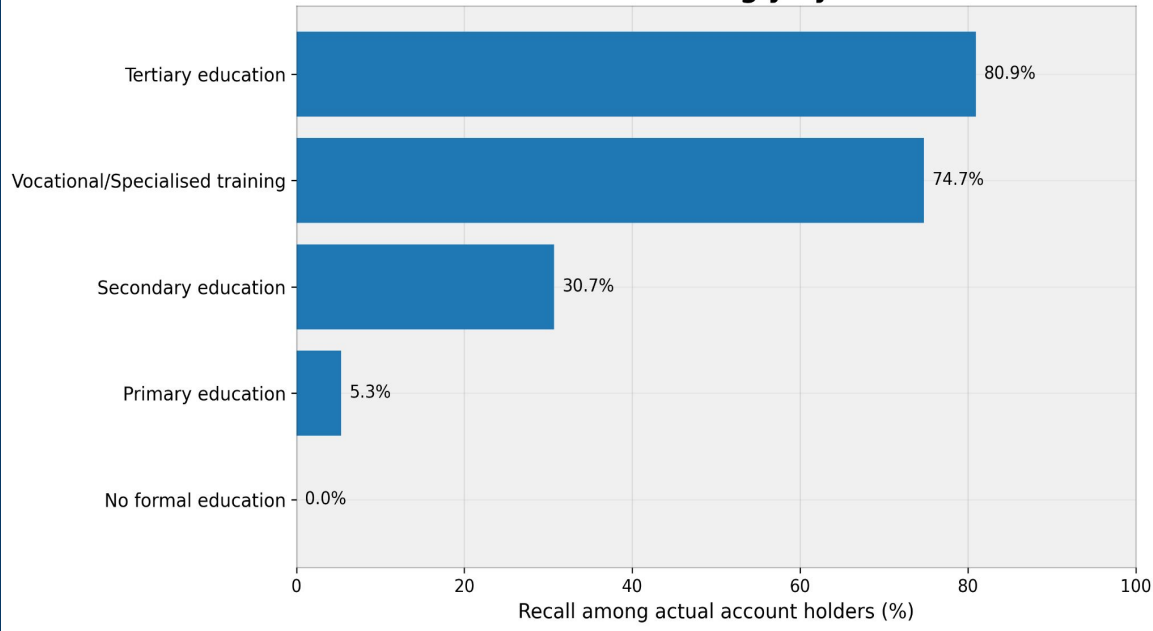


The tuned XGBoost achieves a lower error overall but its errors are concentrated in false negatives

It misses 416 of 662 actual bank-account holders, resulting in a 37.16% recall*

OUR FINDINGS

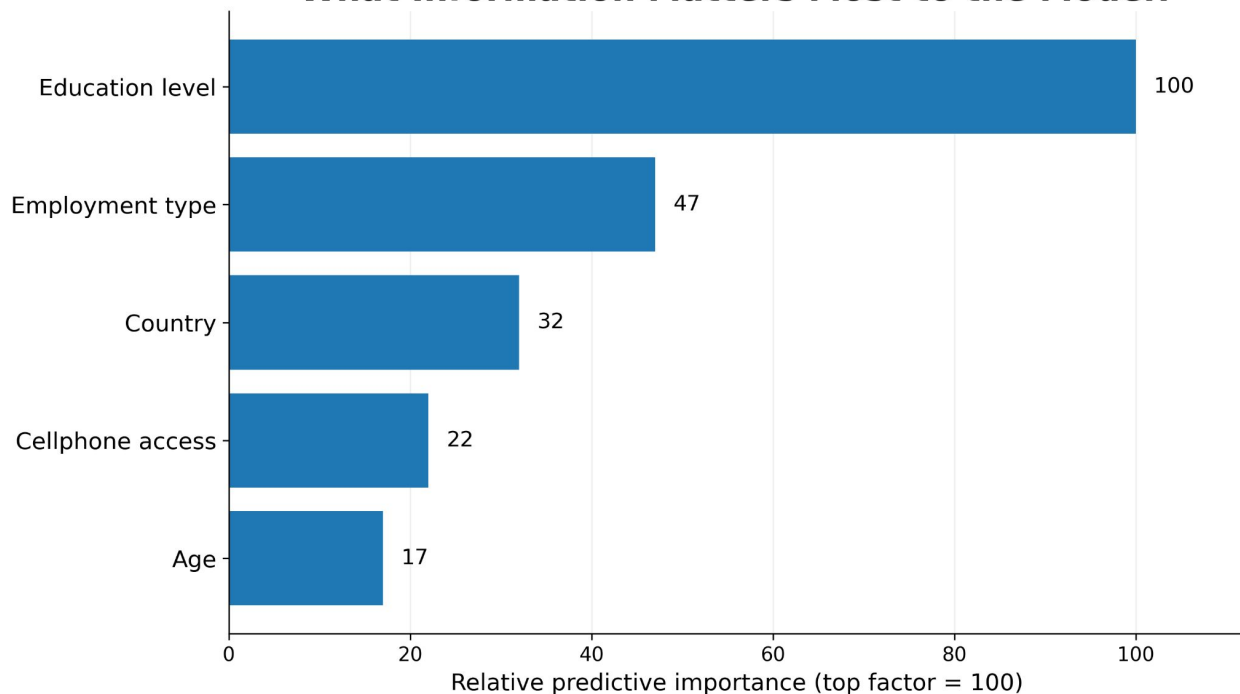
Model Recall Varies Strongly by Education Level



Predictive model performance is highly uneven across education groups: it identifies most tertiary and vocational account holders, but performs very poorly for respondents with primary or no formal education.

OUR FINDINGS

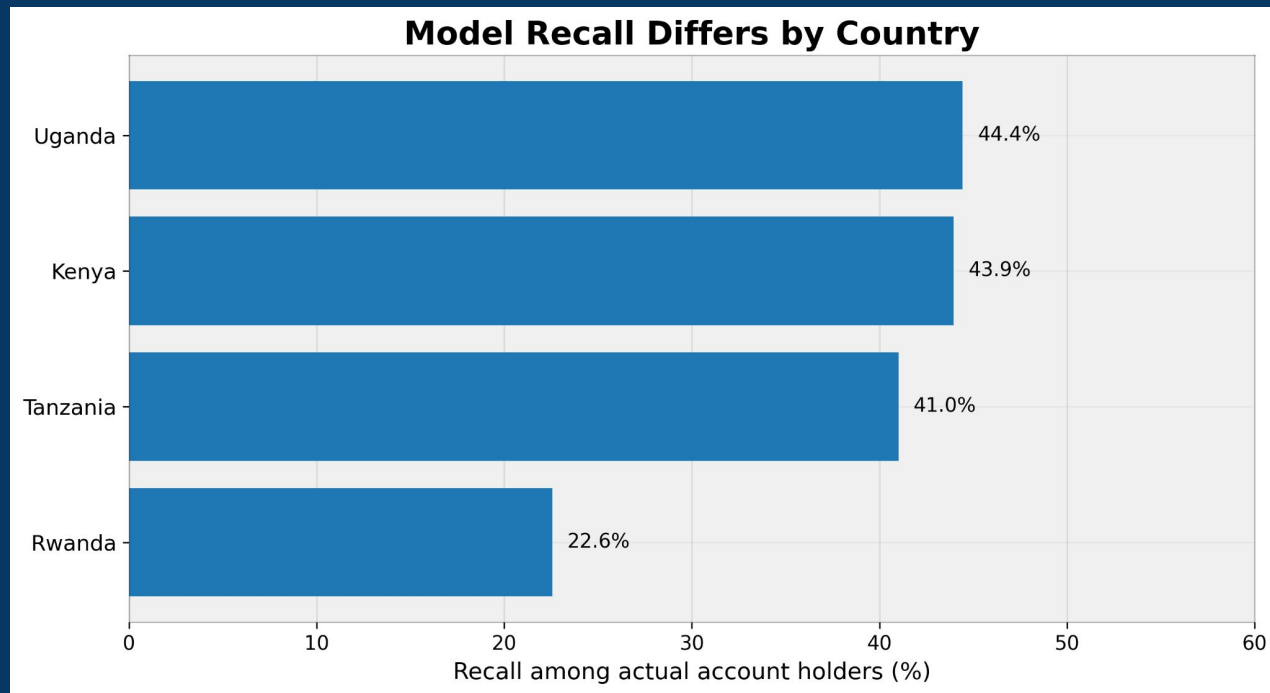
What Information Matters Most to the Model?



The model relies most strongly on education and employment information when distinguishing likely account holders.

Education is by far the strongest predictive signal, with employment type contributing roughly half as much.

OUR FINDINGS



The model identifies account holders at similar rates in Uganda, Kenya and Tanzania, but performs substantially less effectively in Rwanda.

RECOMMENDATIONS

Use the model to focus attention, not make final decisions

Use it to prioritise higher-probability cases for follow-up, while keeping human judgement in the process.

Set the model to match the business priority

If the goal is to identify more potential account holders, accept more false positives. If the goal is fewer false alarms, keep a stricter threshold.

Strengthen performance before scaling

Improve the model for groups where it performs less well especially lower-education and rural populations, and validate performance across countries before wider deployment.

CONCLUSIONS & FUTURE WORK

The selected model correctly classifies about 89% of cases overall.

It is reliable when it identifies an account holder - correct about 72% of the time - but it currently finds only about 37% of all actual account holders. This makes it more suitable for prioritisation than exclusion.

Education, employment, geography and cellphone access provide the strongest predictive signals, but model performance is uneven across population groups.

Next steps: align the model with the business objective, improve coverage of underperforming groups, strengthen the data, and validate performance before wider deployment

THANK YOU
&
QUESTIONS